Dr. Abderraouf Maoudj



Robot learning · Multi-robot coordination · Resilient robotic systems

Postdoctoral Research Fellow & Coordinator of the Robotics Laboratory

Interdisciplinary Research Center for Intelligent Manufacturing and Robotics
King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia

abderraouf.maoudj@kfupm.edu.sa
raouf.maoudj@gmail.com
Google Scholar
amaoudj.github.io
ORCID: 0000-0002-4058-2506
LinkedIn

1,837 citations · h-index 14 · i10-index 16 (Google Scholar, November 2026)

Research Profile

My work builds intelligent robot brains that reason on physical time and act safely on physical robots, with embodied AI for robot arms, legged robots and robot teams as the main application. For more than a decade I have carried out academic and industrial research and development in Algeria, Denmark and Saudi Arabia, with industrial partners that include Siemens, Mobile Industrial Robots and Enabled Robotics. The applications lie in intelligent manufacturing and smart logistics, and in embodied intelligence and the low-altitude economy, which China has named among its future and strategic emerging industries. One question organises the research: how can robots learn useful behaviour from limited interaction, keep working safely when their hardware or their task changes, and coordinate at industrial scale?

- **Embodied AI and world models:** NeuroAct, a robot brain designed to keep working when the body or the task changes, whose shared model generalises in simulation to robot bodies it never saw in training; **multi-agent world models** (current work), in which each robot is curious about how its own actions and those of its teammates change the environment, so that the team learns a joint model of how the whole system evolves; and, as directions I am developing, vision-language-action policies that consult a world model before contact, and policies that reason on physical time.
- **Robot learning for intelligent manufacturing:** as Principal Investigator of KFUPM Grant No. INMR2403, which I proposed, won and delivered, I built a dual-arm Meca500 skill-learning testbed that is now shared laboratory infrastructure in active use. On it, a decentralised policy per arm with graph attention between arms runs on the physical arms and coordinates more arms in simulation than it was trained with. First-author papers on multi-arm reinforcement learning and dual-arm manipulation are in press.
- **Multi-robot systems for smart logistics and drone teams:** decentralised path finding and scheduling for warehouse robot fleets under limited communication, developed with the industrial partner Enabled Robotics, which coordinate very large fleets in simulation; and decentralised multi-drone gas-source localisation with a safety filter, which found the source far more often than infotaxis and model-predictive-control baselines (first author, *Scientific Reports*, accepted 2026, simulation).
- **Fault-tolerant legged robots and actuation:** a co-authored controller in which reinforcement learning modulates central pattern generators was deployed zero-shot on a physical 18-DoF hexapod and kept it walking without a fall across repeated hardware trials, even with multiple legs disabled (under review); the open-source SPINDER hexapod platform (*Scientific Reports*, 2026); and, as Co-Principal Investigator of KFUPM Grant No. INMR2632, a dual-mode actuation joint for an open bipedal platform.

Research Appointments

2023–Present **Postdoctoral Research Fellow & Robotics Laboratory Coordinator**
IRC-IMR, King Fahd University of Petroleum & Minerals, Saudi Arabia
Learning-based manipulation and multi-arm planning; decentralized multi-agent learning; resilient legged control and cross-embodiment policies. PI, INMR2403; Co-PI, INMR2632. Graduate teaching and student supervision.

2021–2023 **Postdoctoral Researcher**
SDU Biorobotics, University of Southern Denmark
Decentralized multi-robot coordination, communication-constrained path finding and capacitated fleet scheduling for industrial logistics. Worked with Enabled Robotics on MiR–UR5 mobile manipulators; Master's supervision.

2018–2020 **Senior Researcher, Robotics Division**
Centre de Développement des Technologies Avancées (CDTA), Algeria
AGV fleet control, intelligent scheduling, collaborative robot integration and Siemens PLC automation for Industry 4.0; engineering teaching and project supervision.

2014–2018 **Robotics Engineer, Robotics Division**
CDTA, Algeria
Robot development, including an industrial AGV and a remotely operated bathymetric survey boat; doctoral research and laboratory teaching at USTHB.

Education

2013–2018 **Ph.D., Process Control and Interactive Rendering**
Université des Sciences et de la Technologie Houari Boumediene (USTHB), Algeria
Distinction: *Très Honorable*. Dissertation: *Teleoperation of Heterogeneous Robots: Application to RobuTER/ULM Robots and AGVM6 Robots.* Advisor: Prof. Brahim Bouzouia.

2010–2012 **M.Sc., Electronics — System Analysis**
Mohammed-Seddik Benyahia University of Jijel, Algeria
Thesis: *Design and Control of a Microcontroller-Based Mobile Robot Prototype.*

2007–2010 **B.Sc., Electronics**
Mohammed-Seddik Benyahia University of Jijel, Algeria

Teaching, Supervision and Mentorship

Current graduate teaching

ICS 520 — Artificial Intelligence and Machine Learning for Robotics, KFUPM, Fall 2026. Autonomous drone applications spanning perception, planning, decision-making and learning.

Courses taught

- **Automation Laboratory (AUTO503)**, undergraduate, USTHB, 2014/2015 and 2015/2016; **Control Laboratory**, Master's in Electronics of Embedded Systems, 2017/2018; **Industrial Programmable Logic Controllers (API)**, Master's in Instrumentation, 2018/2019.
- **Further teaching:** Linear Control Systems; Introduction to Robotic Systems and ROS; Applied Reinforcement Learning for Robotic Manipulation; Embedded Systems for Automotive Engineering; Algorithms and C Programming.
- **Curriculum development:** graduate courses in Reinforcement Learning for Robotics and Sim-to-Real Embodied Robotics, integrating classical control, robot learning and projects on open robotic platforms.

Graduate and project supervision

Co-supervised doctoral students at KFUPM and supervised or co-supervised Master's theses and final-year engineering projects at KFUPM, SDU and CDTA/USTHB. Topics include manipulation, industrial automation, navigation, multi-robot coordination and legged locomotion, with outcomes in peer-reviewed publications, industrial demonstrations and working prototypes.

Research Funding and Leadership

- **Principal Investigator — KFUPM Grant INMR2403 (2023–2025).** *Advancing Robotics for Industry 4.0: Integrating Collaborative Robot Systems to Enhance Industrial Tasks.* Led the project and established the shared dual-arm Meca500 skill-learning testbed. Project record.

- **Co-Principal Investigator — KFUPM Grant INMR2632 (ongoing).** *Intelligent Bipedal Robotic Platform.* Open, modular hardware and control infrastructure for locomotion research.
- **Postdoctoral Researcher — Swarm Robotics for Industry 4.0 (2021–2023).** Independent Research Fund Denmark project at SDU, with Enabled Robotics and IRIDIA, Université libre de Bruxelles. Project record.

Research and Development Projects

Next-Generation Robotic Manipulation and Learning Systems

KFUPM · Ongoing · Postdoctoral Research Fellow

Learning-based manipulation for flexible industrial automation, combining reinforcement learning, imitation learning and inter-arm attention. A graph-attention SAC policy trained with up to four arms coordinates up to ten in simulation and runs on physical Meca500 arms. Current work explores multi-agent world models for sparse-reward coordination and vision–language–action policies for manipulation.

Laboratory demonstrations · Multi-arm planning video

NeuroAct: Transferable Robot Intelligence Across Bodies and Tasks

KFUPM · Ongoing · Postdoctoral Research Fellow

A modular control stack with an embodiment representation, hierarchical policies and persistent memory, designed to reduce retraining for new robots and tasks. A shared model has generalized to unseen synthetic robot morphologies in simulation; physical deployment is the next stage.

Resilient Legged Locomotion Under Progressive Hardware Failures

KFUPM · Ongoing · Postdoctoral Research Fellow

CPG dynamics modulated by reinforcement learning sustain locomotion as multiple joints and legs fail. A single policy trained with randomized fault injection was deployed zero-shot on a physical 18-DoF hexapod, supporting research toward dependable inspection robots. Companion work couples terrain perception and gait adaptation in quadruped simulation.

SPINDER demonstration · Open platform and simulator

Intelligent Bipedal Robotic Platform

KFUPM · Grant INMR2632 · Ongoing · Co-Principal Investigator

Open, modular hardware for reproducible locomotion, balance recovery, jumping and obstacle-negotiation experiments. Replaceable mechanical, sensing and control modules, standardized interfaces and synchronized logging support quantitative comparison of classical and learning-based controllers.

Collaborative Robotics for Industry 4.0

KFUPM · Grant INMR2403 · 2023–2025 · Principal Investigator

Developed multi-cobot skill learning, coordination and motion planning for flexible production, alongside fleet coordination for storage, transport and shipment preparation. Delivered a dual-arm Meca500 testbed with an integrated ROS stack, now used as shared research and teaching infrastructure.

Project record

Swarm Robotics for Industry 4.0

SDU Biorobotics · 2021–2023 · Postdoctoral Researcher

Decentralized planning and scheduling for warehouse fleets operating with limited communication. Delivered capacitated multi-AGV scheduling methods, planners for at least 1,000 simulated robots, and lifelong planning for continuous task arrivals. Co-authored heatmap-guided learning was evaluated with up to 3,500 simulated agents; open planners and datasets support reproducible industrial-logistics research.

Project · ANTS planner · Benchmark dataset · Video

Industry 4.0 Platform Development

CDTA, with Siemens Algeria · 2016–2020 · Robotics Engineer / Senior Researcher

Integrated KUKA LBR iiwa collaborative assembly, RFID part tracking, Siemens PLCs and AGV navigation into a cyber-physical manufacturing platform, connecting robot autonomy with industrial production control.

System demonstration

Aquatic Robot for Bathymetry

CDTA · 2016–2018 · Robotics Engineer

Designed the electronics and control of a remotely operated survey boat delivered to the Algerian National Hydrological Agency, supporting bathymetric measurement and environmental monitoring.

AquaRob publication, IEEE EDiS 2024

Remote Control of Heterogeneous Robots

CDTA · 2014–2016 · Robotics Engineer / Doctoral Researcher

Distributed scheduling and control for heterogeneous industrial robots, reinforcement-learning path optimization and fault-tolerant formation control, supporting coordinated operation of mobile and manipulator teams.

Robots and Systems Designed and Built

- **SPINDER, open-source 18-DoF hexapod:** mechanical design, electronics, firmware, analytic inverse kinematics and hierarchical CPG control. Designs, models and code · Demonstration.
- **Dual-arm Meca500 testbed:** hardware integration and ROS stack for manipulation research and teaching.
- **Industrial AGV:** microcontroller-based motion control and lidar integration.
- **Industry 4.0 assembly cell:** KUKA LBR iiwa, Siemens UHF RFID and PLC integration. Video.
- **Bathymetric survey boat:** electronics and remote control for the Algerian National Hydrological Agency. Publication.

Additional platform experience: UR5, UFactory xArm6 and Lite6, SO-ARM100, MiR mobile robots and Enabled Robotics mobile manipulators.

Technical Expertise

- **Robot learning:** PyTorch; reinforcement learning (PPO, SAC, TD3, DQN, offline RL); multi-agent learning; imitation learning; world models; transformers and vision–language–action models; sim-to-real transfer.
- **Planning and coordination:** multi-agent path finding; multi-arm motion planning; task allocation and scheduling; collision avoidance; mixed-integer optimization (CPLEX), genetic algorithms and vehicle routing.
- **Control and mechatronics:** kinematics and dynamics; nonlinear, fuzzy, sliding-mode and CPG control; control barrier functions; embedded motor control; STM32, Arduino and PIC; robot and sensor integration.
- **Software and automation:** Python, C/C++, MATLAB/Simulink, Java, C# and R; ROS 1/2, MoveIt; MuJoCo, Isaac Lab, PyBullet and CoppeliaSim; Siemens TIA Portal, Ladder and Grafcet; Git, Docker and Linux.

Invited Talks and Research Presentations

- **Reinforcement Learning in Robotics: From Mathematical Foundations to Real-World Applications.** Invited seminar, Electrical Engineering Department, KFUPM, November 2025.
- **Decentralized Multi-Robot Motion Planning and Coordination for Industrial Applications.** Invited seminar, University of Tartu, Estonia, September 2024.
- **Multi-Robot Motion Planning and Coordination in Industrial Environments.** Speaker and co-organizer, Fourth Annual Research Workshop, KFUPM, August 2024.
- Research demonstrations at the KFUPM Research Forum (October 2024); presented research at several international conferences in robotics, automation and industrial informatics.

Professional Service and Industry Engagement

- **Editorial and reviewing:** Topic Editor, *Frontiers in Robotics and AI*, “Human-Robot Interaction in Industrial Settings: New Challenges and Opportunities”; reviewer for robotics and AI journals and conferences.
- **Research infrastructure and open science:** Robotics Laboratory coordination; shared manipulation testbed; open SPINDER hardware and control code, ANTS planner and lifelong multi-agent benchmark datasets.
- **Industrial collaboration:** robotics and automation projects with Siemens Algeria, Enabled Robotics, Mobile Industrial Robots, Sonacome and the Algerian National Hydrological Agency.

Professional References

Prof. Anders Lyhne Christensen
University of Southern Denmark
andc@mmmi.sdu.dk

Prof. Naoyuki Kubota
Tokyo Metropolitan University
kubota@tmu.ac.jp

Prof. Abul Fazal Muhammad Arif
Director, IRC-IMR, KFUPM
abulfazal.arif@kfupm.edu.sa

Dr. Brahim Brahmi
Electrical Engineering / IRC-IMR, KFUPM
brahim.brahmi@kfupm.edu.sa

Publications

Author name shown in bold. Titles link to publisher records where available. Google Scholar.

Journal Articles — Published and Accepted

[J1] A. Aziz Khater, Y. Chen, **A. Maoudj**, A. Aouaichia, "Lyapunov-based adaptive path tracking control of wheeled mobile robots using a self-organizing recurrent Gaussian-gating functional-link fuzzy neural network," *Scientific Reports, published online 1 November 2026*.

[J2] B. Brahmi, **A. Maoudj**, J. Ghommam, "Nonlinear Terminal Model-Based Second-Order Sliding Mode Control for Robust Tracking in Robotic Exoskeletons," *International Journal of Robust and Nonlinear Control, Early View, 30 October 2026*.

[J3] **A. Maoudj**, B. Brahmi, A. A. Saputra, A. Aouaichia, M. Nouioua, N. Kubota, "Decentralized multi-drone gas source localization via belief-coupled MAPPO and barrier-filtered safety," *Scientific Reports*, accepted, 2026.

[J4] **A. Maoudj**, A. A. Saputra, B. Brahmi, M. Nouioua, "SPINDER: An open-source 18-DoF hexapod robot with hierarchical central pattern generator control and analytic inverse kinematics," *Scientific Reports*, vol. 16, art. 26147, 2026.

[J5] **A. Maoudj**, A. A. Saputra, B. Brahmi, N. Kubota, "A hierarchical perception-action framework for adaptive legged locomotion in unstructured environments," *IEEE Access*, vol. 14, pp. 74923–74932, 2026.

[J6] N. Djouambi, M. A. Yallese, M. Kaddeche, **A. Maoudj**, A. Khellaf, M. Nouioua, "Machine-Learning Surrogates for the Multi-Objective Optimisation of Cutting Parameters in Dry Turning of POM C GF25%," *The International Journal of Advanced Manufacturing Technology*, vol. 145, no. 5–6, pp. 3651–3670, 2026.

[J7] M. Nouioua, A. Laouissi, A. A. D. Sarhan, V. Swaminathan, **A. Maoudj**, H. M. Al-Qahtani, "AI-Driven Decision Support for Multi-Objective Optimization of Turning Parameters in Grey Cast Iron Machining," *Results in Engineering*, vol. 29, article 109805, 2026.

[J8] A. Mechta, A. Laouissi, E. Raouache, **A. Maoudj**, M. Nouioua, "Hybrid Numerical-Experimental Optimization of Resistance Spot Welding Parameters for Dissimilar S235/AISI 1075 Steel Sheets via Box–Behnken Response Surface Methodology," *JOM*, vol. 78, no. 11, pp. 10785–10805, 2026.

[J9] Z. Khan, **A. Maoudj**, S. Mekid, "From collaborative robots to Internet of Cobots: Innovations, applications, and emerging trends," *Journal of Ambient Intelligence and Smart Environments*, vol. 18, no. 1, pp. 41–67, 2026.

[J10] M. Boumediene, **A. Maoudj**, A. L. Christensen, "HM-DRL: Enhancing multi-agent pathfinding with a heatmap-based heuristic for distributed deep reinforcement learning," *Applied Intelligence*, vol. 55, no. 12, art. 873, 2025.

[J11] U. Farhadi, H. Hess, **A. Maoudj**, A. L. Christensen, "Evolution of path costs for efficient decentralized multi-agent pathfinding," *Swarm and Evolutionary Computation*, vol. 93, art. 101833, 2025.

[J12] **A. Maoudj**, A. L. Christensen, "Improved decentralized cooperative multi-agent path finding for robots with limited communication," *Swarm Intelligence*, vol. 18, no. 2–3, pp. 167–185, 2024.

[J13] A. Hentout, **A. Maoudj**, A. Kouider, "Shortest path planning and efficient fuzzy logic control of mobile robots in indoor static and dynamic environments," *Romanian Journal of Information Science and Technology*, vol. 27, no. 1, pp. 21–36, 2024.

[J14] **A. Maoudj**, A. Kouider, A. L. Christensen, "The capacitated multi-AGV scheduling problem with conflicting products: Model and a decentralized multi-agent approach," *Robotics and Computer-Integrated Manufacturing*, vol. 81, art. 102514, 2023.

[J15] A. Hentout, **A. Maoudj**, M. Aouache, "A review of the literature on fuzzy-logic approaches for collision-free path planning of manipulator robots," *Artificial Intelligence Review*, vol. 56, no. 4, pp. 3369–3444, 2023.

[J16] A. Hentout, **A. Maoudj**, A. Kouider, M. Aouache, "Effective GA approach for municipal solid waste collection: Dynamic capacitated vehicle routing in smart cities," *International Journal of Artificial Intelligence and Soft Computing*, vol. 7, no. 4, pp. 329–352, 2022.

[J17] **A. Maoudj**, A. Hentout, "Optimal path planning approach based on Q-learning algorithm for mobile robots," *Applied Soft Computing*, vol. 97, art. 106796, 2020.

[J18] A. Hentout, **A. Maoudj**, N. Kaid-Youcef, D. Hebib, B. Bouzouia, "Distributed multi-agent bidding-based approach for the collaborative mapping of unknown indoor environments by a homogeneous mobile robot team," *Journal of Intelligent Systems*, vol. 29, no. 1, pp. 84–99, 2020 (online 2017).

[J19] A. Hentout, M. Aouache, **A. Maoudj**, I. Akli, "Human-robot interaction in industrial collaborative robotics: A literature review of the decade 2008–2017," *Advanced Robotics*, vol. 33, no. 15–16, pp. 764–799, 2019.

[J20] **A. Maoudj**, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, "Distributed multi-agent scheduling and control system for robotic flexible assembly cells," *Journal of Intelligent Manufacturing*, vol. 30, no. 4, pp. 1629–1644, 2019.

[J21] A. Hentout, **A. Maoudj**, M. Ouahal, A. Gasmi, "Fault-tolerant multi-agent scheme for leader/follower formation control of homogeneous mobile robot team," *International Journal of Systems, Control and Communications*, vol. 10, no. 2, pp. 95–125, 2019.

[J22] A. Hentout, **A. Maoudj**, D. Guir, S. Saighi, M. A. Harkat, M. Z. Hammouche, A. Bakdi, "Collision-free path planning for indoor mobile robots based on rapidly-exploring random trees and piecewise cubic Hermite interpolating polynomial," *International Journal of Imaging and Robotics*, vol. 19, no. 3, 2019.

[J23] A. Hentout, **A. Maoudj**, D. Yahiaoui, M. Aouache, "RRT-A*-BT approach for optimal collision-free path planning for mobile robots," *Algerian Journal of Signals and Systems*, vol. 4, no. 2, pp. 39–50, 2019.

[J24] **A. Maoudj**, A. Hentout, B. Bouzouia, R. Toumi, "On-line fault-tolerant fuzzy-based path planning and obstacles avoidance approach for manipulator robots," *International Journal of Uncertainty, Fuzziness and Knowledge-Based Systems*, vol. 26, no. 5, pp. 809–838, 2018.

[J25] A. Bakdi, A. Hentout, H. Boutami, **A. Maoudj**, O. Hachour, B. Bouzouia, "Optimal path planning and execution for mobile robots using genetic algorithm and adaptive fuzzy-logic control," *Robotics and Autonomous Systems*, vol. 89, pp. 95–109, 2017.

Refereed Conference Papers

[C1] **A. Maoudj**, A. Kharidege, B. Brahmi, A. A. Saputra, M. Nouioua, "An Adaptive Vision-Based Robotic System for Flexible Surface Polishing," *23rd International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, vol. 2, pp. 1091–1098, 2026.

[C2] B. Brahmi, **A. Maoudj**, J. Ghommam, M. H. Rahman, "Hamiltonian-Based Human Intention Estimation for Compliant Exoskeleton Robot," *23rd International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, vol. 1, pp. 495–502, 2026.

[C3] K. Amara, W. Dib, A. O. Djekoune, **A. Maoudj**, N. Zenati, "AquaRob: An unmanned surface drone for the coastal surveillance and environmental monitoring," *Proc. 4th International Conference on Embedded & Distributed Systems (EDiS), IEEE*, pp. 193–198, 2024.

[C4] **A. Maoudj**, S. Mekid, A. L. Christensen, "Lifelong MAPF for shipment preparation in stochastic warehouses," *Proc. IEEE 20th International Conference on Automation Science and Engineering (CASE)*, pp. 1006–1011, 2024.

[C5] **A. Maoudj**, A. L. Christensen, "Decentralized multi-agent path finding in dynamic warehouse environments," *Proc. 21st International Conference on Advanced Robotics (ICAR), IEEE*, pp. 28–34, 2023.

[C6] **A. Maoudj**, A. L. Christensen, "Decentralized multi-agent path finding in warehouse environments for fleets of mobile robots with limited communication range," *Swarm Intelligence: 13th International Conference (ANTS 2022), Springer LNCS vol. 13491*, pp. 104–116, 2022.

[C7] M. Gaham, M. Kadri, **A. Maoudj**, A. Bendjelloul, A. Kouider, B. Mihoubi, K. Tebani, B. Bouzouia, "Multi-agent control of an Industry 4.0 cyber-physical assembly cell," *1st National Conference on Applied Computing and Smart Technologies, Sidi Bel Abbès, Algeria*, 2021.

[C8] **A. Maoudj**, A. Hentout, A. L. Christensen, A. Kouider, "A fast constructive path planning algorithm for mobile robot navigation," *Proc. 26th IEEE International Conference on Emerging Technologies and Factory Automation (ETFA)*, pp. 1–8, 2021.

[C9] **A. Maoudj**, A. L. Christensen, "Q-learning-based navigation for mobile robots in continuous and dynamic environments," *Proc. IEEE 17th International Conference on Automation Science and Engineering (CASE)*, pp. 1338–1345, 2021.

[C10] **A. Maoudj**, A. Hentout, M. Bouhebila, C. Boubakir, H. Maoudj, "Design and control of an educational and research-oriented mobile robot," *International Conference on Networking, Telecommunications, Biomedical Engineering and Applications (ICNTBA)*, 2019.

[C11] H. Maoudj, **A. Maoudj**, A. Hentout, M. Bouhebila, C. Boubakir, "Autonomous fuzzy-based navigation of mobile robots: Design, realization and control of a prototype," *International Conference on Networking, Telecommunications, Biomedical Engineering and Applications (ICNTBA)*, 2019.

[C12] A. Hentout, A. Tiberkak, **A. Maoudj**, L. Berkat, A. Abeb, "Virtual pheromone-based approach for objects searching in RFID-based cyber-physical robotic systems," *Proc. International Conference on Applied Smart Systems (ICASS)*, pp. 1–7, 2018.

[C13] A. Hentout, A. Beghni, A. B. Nourine, **A. Maoudj**, B. Bouzouia, "Mobile robot RBPF-SLAM with LMS sensor in indoor environments: Application on RobuTER robot," *Proc. 5th International Conference on Electrical Engineering, Boumerdes (ICEE-B), IEEE*, 2017.

[C14] A. Hentout, **A. Maoudj**, B. Bouzouia, "A survey of development frameworks for robotics," *Proc. 8th International Conference on Modelling, Identification and Control (ICMIC)*, pp. 67–72, 2016.

[C15] **A. Maoudj**, B. Bouzouia, A. Hentout, A. Kouider, R. Toumi, "Distributed multi-agent approach based on priority rules and genetic algorithm for tasks scheduling in multi-robot cells," *Proc. IECON 2016, 42nd Annual Conference of the IEEE Industrial Electronics Society*, pp. 692–697, 2016.

[C16] **A. Maoudj**, B. Bouzouia, A. Hentout, R. Toumi, "Multi-agent approach for task allocation and scheduling in cooperative heterogeneous multi-robot team: Simulation results," *Proc. IEEE 13th International Conference on Industrial Informatics (INDIN)*, pp. 179–184, 2015.

[C17] **A. Maoudj**, B. Bouzouia, A. Hentout, R. Toumi, "Multi-agent architecture for telerobotics of heterogeneous multi-robot cooperation through negotiated task allocation with constraints: Initial implementation," *1re Journée d'Étude en Automatique et ses Applications (1JEAA14)*, pp. 38–43, 2014.

Editorial, Workshop Paper, Preprints and Theses

[O1] S. Bouraine, M. Ammi, K. Geihs, A. Hentout, **A. Maoudj**, F. Yacef, "Editorial: Human-robot interaction in industrial settings: new challenges and opportunities," *Frontiers in Robotics and AI*, vol. 12, art. 1652426, 2025.

[O2] A. Hentout, M. Aouache, **A. Maoudj**, I. Akli, "Key challenges and open issues of industrial collaborative robotics," *Workshop on Human-Robot Interaction: From Service to Industry (HRI-SI), held at IEEE RO-MAN*, 2018.

[O3] A. Kharidege, **A. Maoudj**, "Integration of a vision system with a robotic polishing system for machining variety of workpieces," *Research Square preprint*, 2024.

[O4] A. Hentout, **A. Maoudj**, A. Kouider, "Efficient fuzzy-logic control of indoor mobile robots evolving in static and dynamic workspaces," *Research Square preprint*, 2023.

[O5] A. Hentout, **A. Maoudj**, A. Kouider, "A literature review of service robotics: Challenges and open research issues," *Workshop on Human-Robot Interaction: From Service to Industry (HRI-SI)*, 2022.

[O6] A. Hentout, I. Akli, **A. Maoudj**, A. Tiberkak, "Developing a distributed architecture for robots control in industrial RFID-based cyber-physical systems," *technical report/preprint*, 2018.

[O7] **A. Maoudj**, "Téléopération multi-robots hétérogènes : Application au cas des robots RobuTER/ULM et AGV-M6," *Doctoral thesis, USTHB*, 2018.

[O8] **A. Maoudj**, C. E. Boubakir, "Conception et commande d'un prototype mobile à base de microcontrôleur," *Master's thesis, Mohammed-Seddik Benyahia University of Jijel*, 2012.

Journal Manuscripts — Under Review and Submitted

[U1] A. A. Saputra, **A. Maoudj**, M. L. Afakh, A. R. Ahmad, S. Mekid, N. Kubota, "Decentralized RL-CPG control for hexapod locomotion under simultaneous multi-leg failures," *IEEE Transactions on Robotics*, under review, 2026.

[U2] **A. Maoudj**, S. Mekid, M. Nouioua, "ML-BiRRT: Machine-learning-enhanced bi-directional RRT for multi-arm robot motion planning and coordination," *Robotics and Autonomous Systems*, under review, 2026.

[U3] **A. Maoudj**, A. A. Khater, B. Brahmi, A. Aouaichia, A. M. Sallam, A. A. Saputra, "Online fuzzy-neural actor-critic control for hexapod trajectory tracking," *Applied Soft Computing*, under review, 2026.

[U4] M. Boumediene, A. L. Christensen, **A. Maoudj**, "MAPF-Relocate: Mitigating congestion through strategic agent relocation in multi-agent pathfinding," *Journal of Intelligent & Robotic Systems*, under review, 2026.

[U5] **A. Maoudj**, A. A. Khater, B. Brahmi, A. Aouaichia, A. M. Sallam, A. A. Saputra, P. K. M. Parameswarappa, "Lyapunov-stable path tracking control of wheeled mobile robots using self-organizing recurrent probabilistic functional-link fuzzy neural network," *Biomimetic Intelligence and Robotics*, under review, 2026.

[U6] B. Brahmi, **A. Maoudj**, N. Iqbal, M. H. Rahman, "Model-based terminal second-order sliding mode control with exact finite-time convergence for fault-tolerant exoskeleton robots," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, under review, 2026.

[U7] A. A. Khater, A. Aouaichia, **A. Maoudj**, Y. Chen, "An advanced recurrent probabilistic TSK neuro-fuzzy identifier for mobile robot system," *Engineering Applications of Artificial Intelligence*, under review, 2026.

[U8] S. Alqahtani, **A. Maoudj**, S. Bashmal, "Learning-based kinematic modeling and disturbance compensation for a 3UPU parallel kinematic manipulator under base vibrations," *submitted*, 2026.